Seara Data

Seara, Menemani Perjalanan
Datamu Lebih Terarah.

Portfolio Bootcamp Data Analyst – SQL

Batch 2

By Rahmanida S

SEARA DATA

Seara, Menemani Perjalanan
Datamu Lebih Terarah.

Bootcamp Data Analyst

Tools Wajib Data Analyst

Excel

Power BI

Python

SQL

4 Weeks Intensive Bootcamp

SEARA

Seara, Menemani Perjalanan
Datamu Lebih Terarah.

About Me

As a Chemistry graduate and an experienced QA/QC Supervisor with a four-year track record in manufacturing, I focus on driving process optimization and problem-solving. In today's landscape, I observe that the most impactful operational decisions are entirely data-driven. This insight inspired me to develop expertise in data analytics. To leverage this expertise and expand my career reach toward strategic analytical roles, I am committed to building new technical capabilities through this Data Analyst Bootcamp.

Positioning myself for a strategic transition into data analytics, I am continuously building technical skills in advanced data processing tools. I aim to merge my industrial experience with these new frameworks to architect smarter, high-impact business decisions.

Data Analyst Learning Journey

Excel

Understand & Prepare Data
Become a Data Analyst
Microsoft Excel Basic
Microsoft Excel Intermediate
Power Query

Output: Clean dataset ready for analysis.

Power BI

Analyze & Visualize Data
Microsoft Power BI Basic
DAX & Data Calculation
Microsoft Power BI Advanced
Visualization & Interactive Report

Output: Interactive dashboard and business insights.

Python

Transform & Automate Data
Python Basic
Data Cleaning & Formatting
Python Advanced
Data Transformation

Output: Processed data pipeline and efficient workflows.

SQL

Query & Extract Insights
SQL Basic
Query & Filter Data
SQL Intermediate
Join, Aggregation & Subquery

Output: Insights directly from databases.

Summary of SQL Session

SEARA DATA

Seara, Menemani Perjalanan Datamu Lebih Terarah.

What I Learned

1 Data Fundamentals
- Schema & data types
- Data period & partition
- Understand data grain
- NULL / nullable fields
- Preview data before query

2 Query & Transform
- SELECT, FROM, WHERE
- ORDER BY & LIMIT
- FILTER BY DATE
- IS NULL / COALESCE
- CTE with WITH

3 Aggregation
- COUNT, SUM, AVG, MIN, MAX
- GROUP BY
- WHERE vs HAVING
- Define Dimensions & Metrics
- Business metrics

4 JOIN & Validation
- INNER JOIN, LEFT JOIN
- Join keys & cardinality
- Detect fanout
- Validate rows before & after JOIN

5 Advanced Analysis
- Window functions
- ROW_NUMBER() for ranking
- SUM() OVER for contribution
- LAG() for comparison
- Qualify for filtering windows result

6 Insight & Action
- Turn analysis into findings
- Compare results across periods
- Explain what the numbers mean
- Give a clear recommendation
- Use simple visuals
- State limitations

7 SQL & BigQuery Habits
- Select only needed columns
- Filter partition dates
- Check bytes scanned
- Use readable CTEs
- Verify before sharing

Summary of SQL Session

SEARA DATA

Seara, Menemani Perjalanan Datamu Lebih Terarah.

What I Learned

SQL Analysis: End-to-End Workflow

1 — Business Questions

- Understand the business brief
- Clarify the problem & objective
- Define key questions
- Set metrics and scope

Goal: Clear direction

2 — Data Profiling

- Check schema & data types
- Review data period & grain
- Check NULLs / completeness
- Identify partition & structure

Goal: Trustworthy data understanding

3 — Query & Aggregate

- Extract and filter data
- Build metrics with aggregations
- Use GROUP BY & HAVING
- Apply CASE WHEN
- CTEs

Goal: Accurate metrics

4 — JOIN & Analyze

- Combine datasets
- Check join cardinality
- Validate fanout & totals
- Use window functions

Goal: Reliable analysis

5 — Visualize & Share Insights

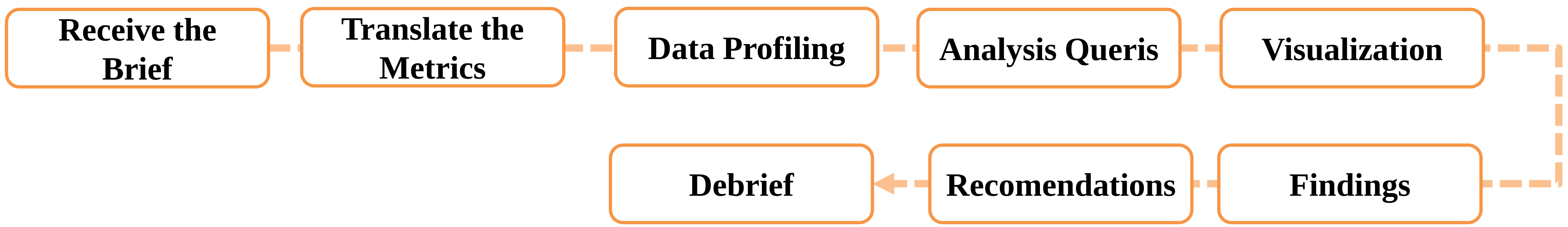
- Build one clear visualization
- Deliver 3 key findings
- Provide 1 recommendation
- State limitations

Goal: Better business decisions

Bussines Question?

“December sales were much higher than November, so why did profit go down according to the Finance team?”

Work Flow



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- Script
- Visualization
- Analysis Data
- Key Insight
- Link GitHub

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Metrics Bussines

Business Question	Metric
Is sales performance strong?	Gross Revenue
Did sales increase in December?	MoM Growth
Did profit decline?	Gross Profit
What might be putting pressure on profit?	Discounts / COGS / Product Mix / Channel

Brief Definition	Definition Used
Strong Sales	Gross Revenue = SUM(Qty × Unit Price)
Increase Compared to November	MoM Growth using LAG on monthly aggregates
Profit Decline	Gross Profit = Net Revenue − (Qty × Unit Cost)
Period Coverage	December 2025, compared with November 2025 and Q3 2025
Channel Coverage	All channels, as no specific channel limitation was stated in the brief

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Data Profiling

1. Dataset & Table on BigQuery

data-analyst-seara-data / Datasets / SearaData

SearaData — Create Table

Overview | Details | Insights

Tables | Graphs | Models | Routines

Filter — Enter property name or value

Table ID	Type	Create time
Pelanggan	Table	Aug 18, 2026, 8:30:05 PM UTC+7
Produk	Table	Aug 18, 2026, 8:29:20 PM UTC+7
Toko	Table	Aug 18, 2026, 8:37:23 PM UTC+7
Tranksaksi	Table	Aug 18, 2026, 8:27:50 PM UTC+7

Row	tanggal	order_id	pelanggan_id	toko_id
1	2024-02-13	ORD-2024-000435	P02771	T06
2	2024-04-29	ORD-2024-001038	P00917	T02
3	2025-03-31	ORD-2025-004088	P03420	T05
4	2025-06-05	ORD-2025-004934	P00182	T11
5	2025-06-25	ORD-2025-005167	P02818	T02
6	2025-10-16	ORD-2025-006294	P02481	T05
7	2025-10-28	ORD-2025-006150	P01258	T10
8	2025-11-02	ORD-2025-006620	P02186	T08
9	2025-11-17	ORD-2025-006523	P00843	T05

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Data Profiling

2. Profiling Data

Data Latihan untuk Por... Run Save

File Edit Run Tools

```
#. Profiling Data
SELECT
  COUNT(*) AS total_baris,
  COUNT(DISTINCT order_id) AS total_order,
  MIN(tanggal) AS tanggal_awal,
  MAX(tanggal) AS tanggal_akhir
FROM `data-analyst-seara-data.SearaData.Tranksaksi`
```

Result Preview

Job information | Results | Visualization | JSON | Execution detai

Row	total_baris	total_order	tanggal_awal	tanggal_akhir
1	10970	7323	2024-01-01	2025-12-31

Check	Result
Total Rows	10,970
Total Unique Orders	7,323
Earliest Date	January 1, 2024
Latest Date	December 31, 2025

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Data Profiling

2. Profiling Data

```sql
#. Cek Missing Value dan Data Tidak Valid
SELECT
  COUNTIF(tanggal IS NULL) AS tanggal_null,
  COUNTIF(order_id IS NULL) AS order_null,
  COUNTIF(qty IS NULL) AS qty_null,
  COUNTIF(harga_satuan IS NULL) AS harga_null,
  COUNTIF(qty <= 0) AS qty_tidak_valid
FROM `data-analyst-seara-data.SearaData.Tranksaksi`;
```

Result Preview

Job information | Results | Visualization | JSON | Execution details | Execution gr

Row	tanggal_null	order_null	qty_null	harga_null	qty_tidak_valid
1	0	0	0	0	0

Check	Result	Conclusion
Total Rows	10,970	Data contains 10,970 rows
Total Unique Orders	7,323	There are 7,323 unique orders
Period	2024-01-01 to 2025-12-31	Data covers 2 years
tanggal NULL	0	☑ Data is clear
order_id NULL	0	☑ Data is clear
qty NULL	0	☑ Data is clear
harga_satuan NULL	0	☑ Data is clear
qty <= 0	0	☑ Data is clear

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Data Analysis

1. Gross Revenue Total

```sql
#1.Mencari Angka Total Gross Revenue

SELECT
  SUM(qty * harga_satuan) AS total_gross_revenue
FROM `data-analyst-seara-data.SearaData.Tranksaksi`
WHERE tanggal >= DATE '2025-01-01'
  AND tanggal <= DATE '2025-12-31';
```

```sql
  #. Pecah Gross Revenue Per-Bulan
SELECT
  DATE_TRUNC(tanggal, MONTH) AS bulan,
  SUM(qty * harga_satuan) AS gross_revenue
FROM `data-analyst-seara-data.SearaData.Tranksaksi`
WHERE tanggal >= DATE '2025-01-01'
  AND tanggal <= DATE '2025-12-31'
GROUP BY bulan
ORDER BY bulan;
```

Result Preview

Job information | Results | Visualization | JSON

Row	total_gross_reven...
1	27307690000

Job information | Results | Visualization | JSON

Row	bulan	gross_revenue
1	2025-01-01	1988000000
2	2025-02-01	1829100000
3	2025-03-01	2132020000
4	2025-04-01	2564200000

Metric	Result
Total Gross Revenue	Rp27,307,690,000
Sum of Gross Revenue (All Months)	Rp27,307,690,000
Validation	☑ Same Result → **Data is Valid**

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Data Analysis

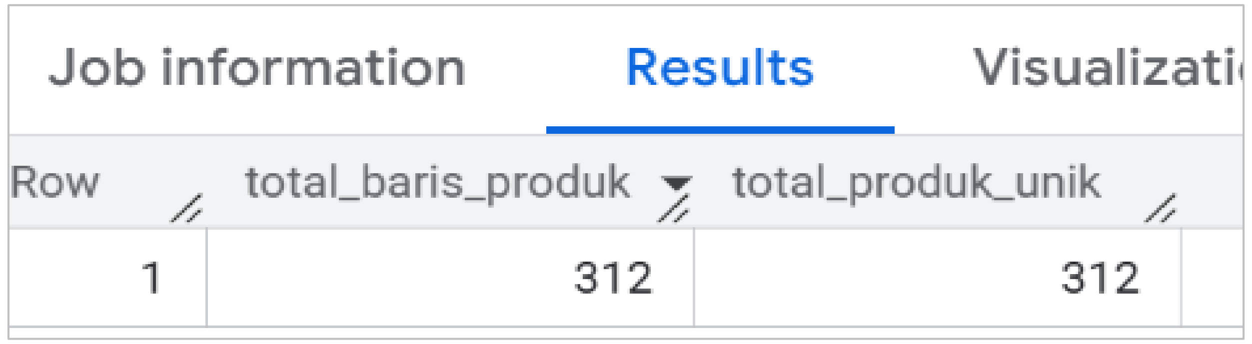
2. Check Duplicate Product_id Before JOIN

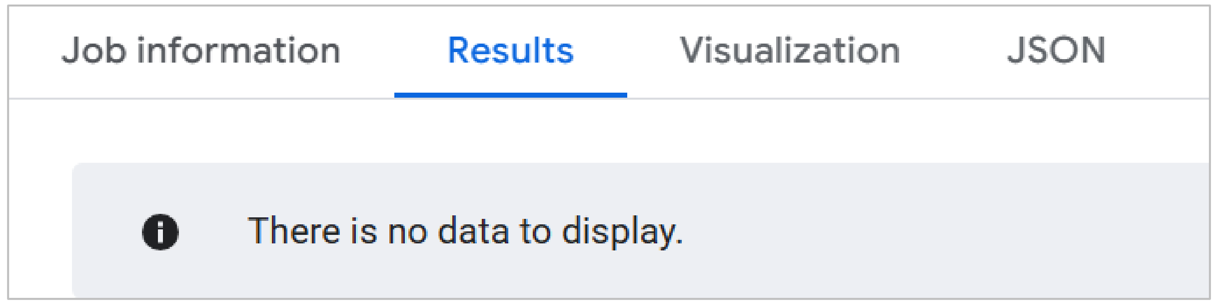
```sql
#. Cek Kode Unik (Product_id)
SELECT
  COUNT(*) AS total_baris_produk,
  COUNT(DISTINCT produk_id) AS total_produk_unik
FROM `data-analyst-seara-data.SearaData.Produk`;

#. Cek Duplikasi Product_id

SELECT
  produk_id,
  COUNT(*) AS jumlah
FROM `data-analyst-seara-data.SearaData.Produk`
GROUP BY produk_id
HAVING COUNT(*) > 1
ORDER BY jumlah DESC;
```

Result Preview





Check	Result
Total Products	312
Unique produk_id	312
Conclusion	No duplicate produk_id found in the Product table.

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Data Analysis

2. JOIN DATA 'TRANSAKSI' WITH 'PRODUK' TABLE

```
#. Join Data Transaksi dengan Produk
SELECT
  t.order_id,
  t.tanggal,
  t.produk_id,
  t.qty,
  t.harga_satuan,
  p.harga_modal
FROM `data-analyst-seara-data.SearaData.Tranksaksi` AS t
INNER JOIN `data-analyst-seara-data.SearaData.Produk` AS p
  ON t.produk_id = p.produk_id
WHERE t.tanggal >= DATE '2025-01-01'
  AND t.tanggal <= DATE '2025-12-31'
ORDER BY t.tanggal;
```

3. COGS TOTAL

```
#. Hitung HPP Total 2025

SELECT
  SUM(t.qty * p.harga_modal) AS total_hpp
FROM `data-analyst-seara-data.SearaData.Tranksaksi` AS t
INNER JOIN `data-analyst-seara-data.SearaData.Produk` AS p
  ON t.produk_id = p.produk_id
WHERE t.tanggal >= DATE '2025-01-01'
  AND t.tanggal <= DATE '2025-12-31';
```

4. VALIDATION AFTER JOIN DATA

```
  #. Validasi Data After Join
  SELECT
  COUNT(*) AS total_baris_setelah_join,
  COUNT(DISTINCT t.order_id) AS total_order_setelah_join,
  SUM(t.qty * t.harga_satuan) AS gross_revenue_setelah_join,
  SUM(t.qty * p.harga_modal) AS hpp_setelah_join
FROM `data-analyst-seara-data.SearaData.Tranksaksi` AS t
INNER JOIN `data-analyst-seara-data.SearaData.Produk` AS p
  ON t.produk_id = p.produk_id
WHERE t.tanggal >= DATE '2025-01-01'
  AND t.tanggal <= DATE '2025-12-31';
```

Metric	Result
Rows After JOIN	6,326
Unique Orders After JOIN	3,973
Gross Revenue After JOIN	Rp27,307,690,000
COGS After JOIN	Rp17,555,610,000
Gross Revenue Before JOIN	Rp27,307,690,000
Gross Revenue After JOIN	**Rp27,307,690,000**
Validation	☑ Gross Revenue remains the same → **Data is valid**

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Data Analysis

5. Breakdown Gross Revenue, Net Revenue After Discount, COGs, Profit by Monthly

```
SELECT
  DATE_TRUNC(t.tanggal, MONTH) AS bulan,

  -- Gross Revenue
  SUM(t.qty * t.harga_satuan) AS gross_revenue

  -- Net Revenue setelah diskon
  SUM(
    t.qty * t.harga_satuan
    * (1 - t.diskon_pct / 100)
  ) AS net_revenue,

  -- HPP
  SUM(t.qty * p.harga_modal) AS hpp,
```

```
  -- Gross Profit
  SUM(
    t.qty * t.harga_satuan
    * (1 - t.diskon_pct / 100)
  )
  - SUM(t.qty * p.harga_modal) AS gross_profit

FROM `data-analyst-seara-data.SearaData.Tranksaksi` AS t

INNER JOIN `data-analyst-seara-data.SearaData.Produk` AS p
  ON t.produk_id = p.produk_id

WHERE t.tanggal >= DATE '2025-01-01'
  AND t.tanggal <= DATE '2025-12-31'

GROUP BY bulan
ORDER BY bulan;
```

Metric	November	December
Gross Revenue	Rp2,538,670,000	Rp3,183,260,000
Net Revenue	Rp2,052,268,800	Rp2,390,667,420
COGS	Rp1,630,300,000	Rp2,040,970,000
Gross Profit	Rp421,968,800	Rp349,697,420

Data Analysis

6. MoM

```
# MoM Revenue

WITH per_bulan AS (
  SELECT
    DATE_TRUNC(t.tanggal, MONTH) AS bulan,

    SUM(t.qty * t.harga_satuan) AS gross_revenue,

    SUM(
      t.qty * t.harga_satuan
      * (1 - t.diskon_pct / 100)
    ) AS net_revenue,

    SUM(t.qty * p.harga_modal) AS hpp,
```

```
  SUM(
    t.qty * t.harga_satuan
    * (1 - t.diskon_pct / 100)
  ) - SUM(t.qty * p.harga_modal) AS gross_profit

FROM `data-analyst-seara-data.SearaData.Tranksaksi` AS t

INNER JOIN `data-analyst-seara-data.SearaData.Produk` AS p
  ON t.produk_id = p.produk_id

WHERE t.tanggal >= DATE '2025-01-01'
  AND t.tanggal <= DATE '2025-12-31'

GROUP BY bulan
```

7. Data Analysis (November, Desember & Quarter)

Analysis	Comparison	Result
MoM Revenue	Dec vs Nov	+25.4%
MoM Gross Profit	Dec vs Nov	-17.1%
Quartal (QoQ) Gross Profit	Q4 vs Q3	-25.7%

Quarter	Net Revenue	COGs	Gross Profit
Q1	Rp5,410 M	Rp3,825 M	Rp1,585 M
Q2	Rp5,879 M	Rp4,193 M	Rp1,686 M
Q3	**Rp5,985 M**	**Rp4,238 M**	**Rp1,747 M**
Q4	**Rp6,597 M**	**Rp5,298 M**	**Rp1,299 M**

Data Analysis

7. Data Analysis

Quarter	Net Revenue	HPP	Gross Profit	QoQ Profit
Q1	Rp5,410 M	Rp3,825 M	Rp1,585 M	—
Q2	Rp5,879 M	Rp4,193 M	Rp1,686 M	+6,4%
Q3	Rp5,985 M	Rp4,238 M	**Rp1,747 M**	**+3,6%**
Q4	Rp6,597 M	Rp5,298 M	**Rp1,299 M**	**-25,7%**

8. Data Analysis (What Happened from Q3 → Q4?)

Metric	Q3	Q4	Change
Net Revenue	Rp5.985M	Rp6.597M	+10.2%
COGS	Rp4.238M	Rp5.298M	+25.0%
Gross Profit	Rp1.747M	Rp1.299M	-25.7%
Gross Margin	29.2%	19.7%	-9.5 pts

Key Insight:

Revenue **increased by 10.2%**, but COGS **increased by 25.0%**.

This means the **cost of goods sold increased much faster than revenue**. **As a result**, Gross Profit **decreased by 25.7%.**

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Documentations & Script

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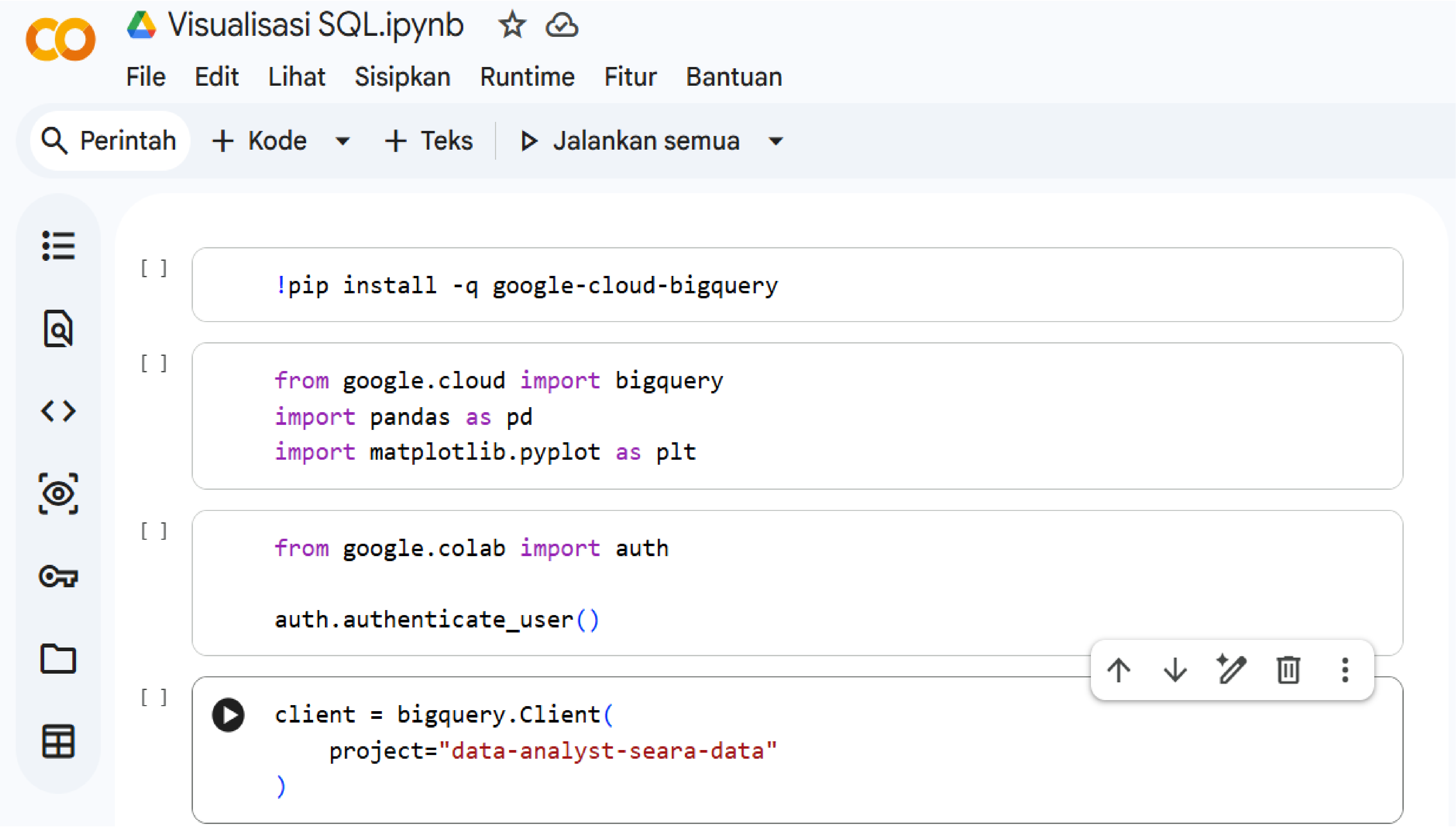
Data Analysis

9. Visualization Using Python (Google Collabs)




```
!pip install -q google-cloud-bigquery
```

```
from google.cloud import bigquery
import pandas as pd
import matplotlib.pyplot as plt
```

```
from google.colab import auth

auth.authenticate_user()
```

```
client = bigquery.Client(
    project="data-analyst-seara-data"
)
```

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Visualization

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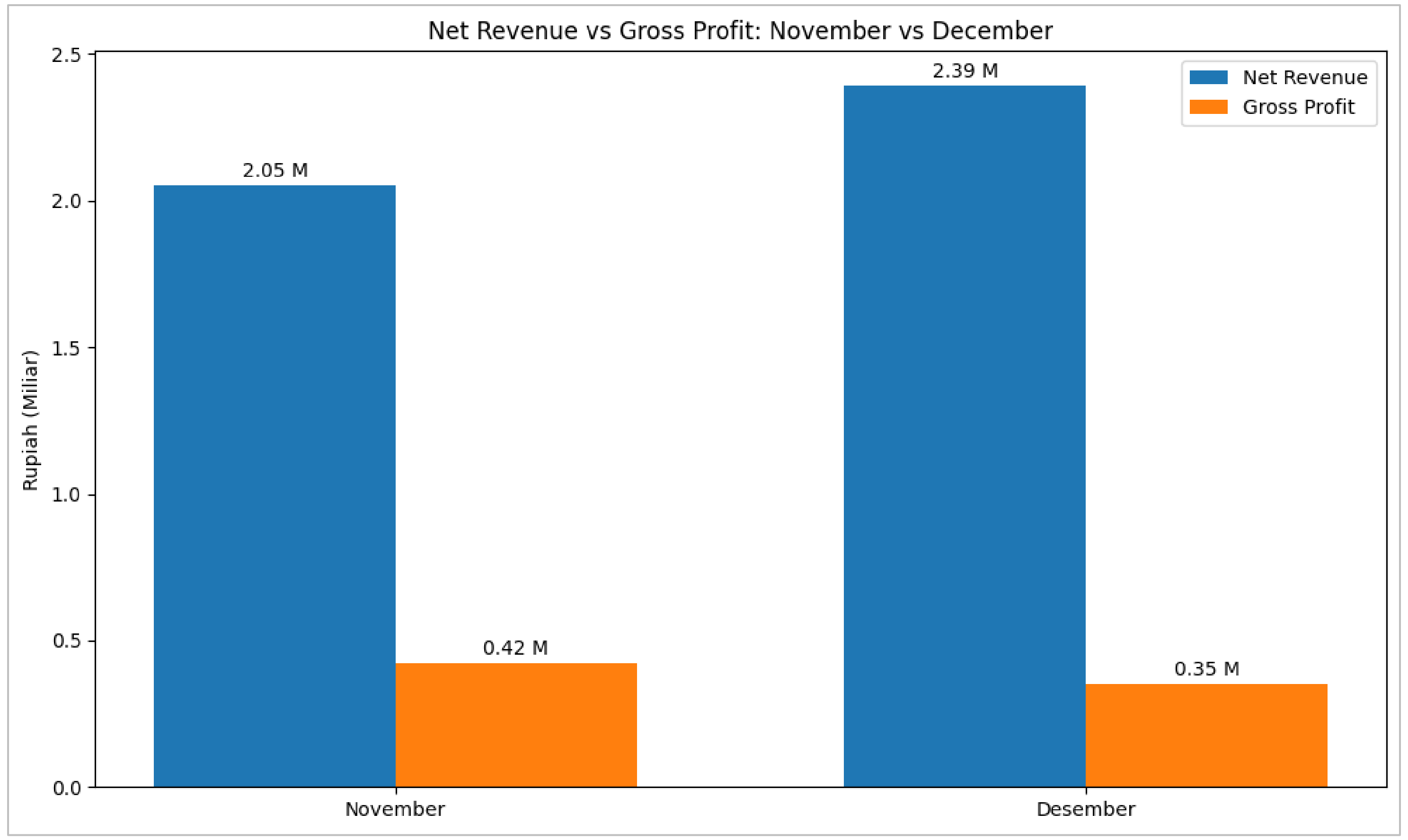
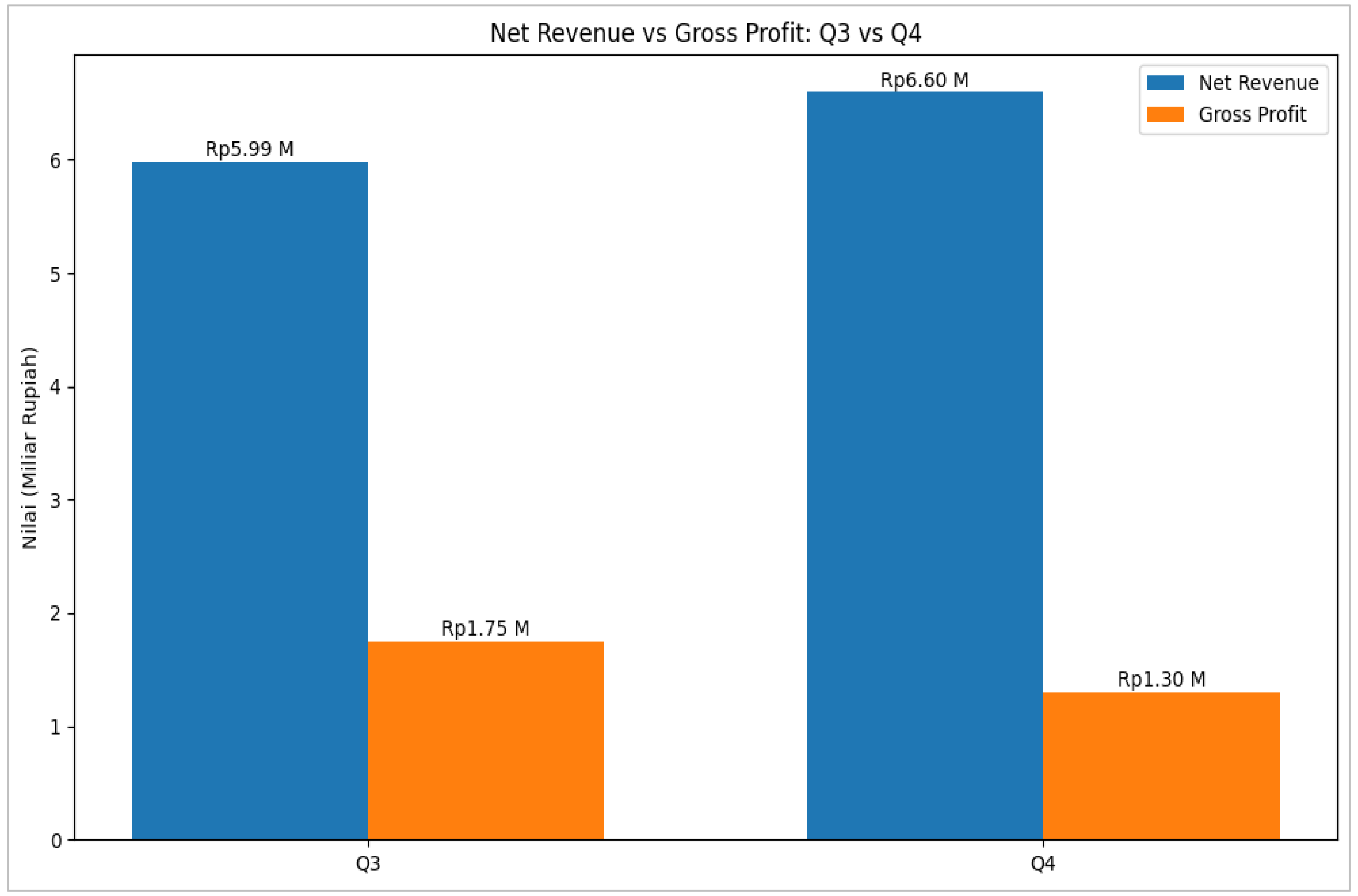
Data Analysis

Questions : Did December sales really increase compared to November, and how did profit perform?





Answer:
Based on the data, revenue increased, but profit actually declined—revenue rose by 16.5% in December vs. November and by 10.2% in Q4 vs. Q3, while gross profit fell by 17.1% and 25.7%, respectively.

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Visualization

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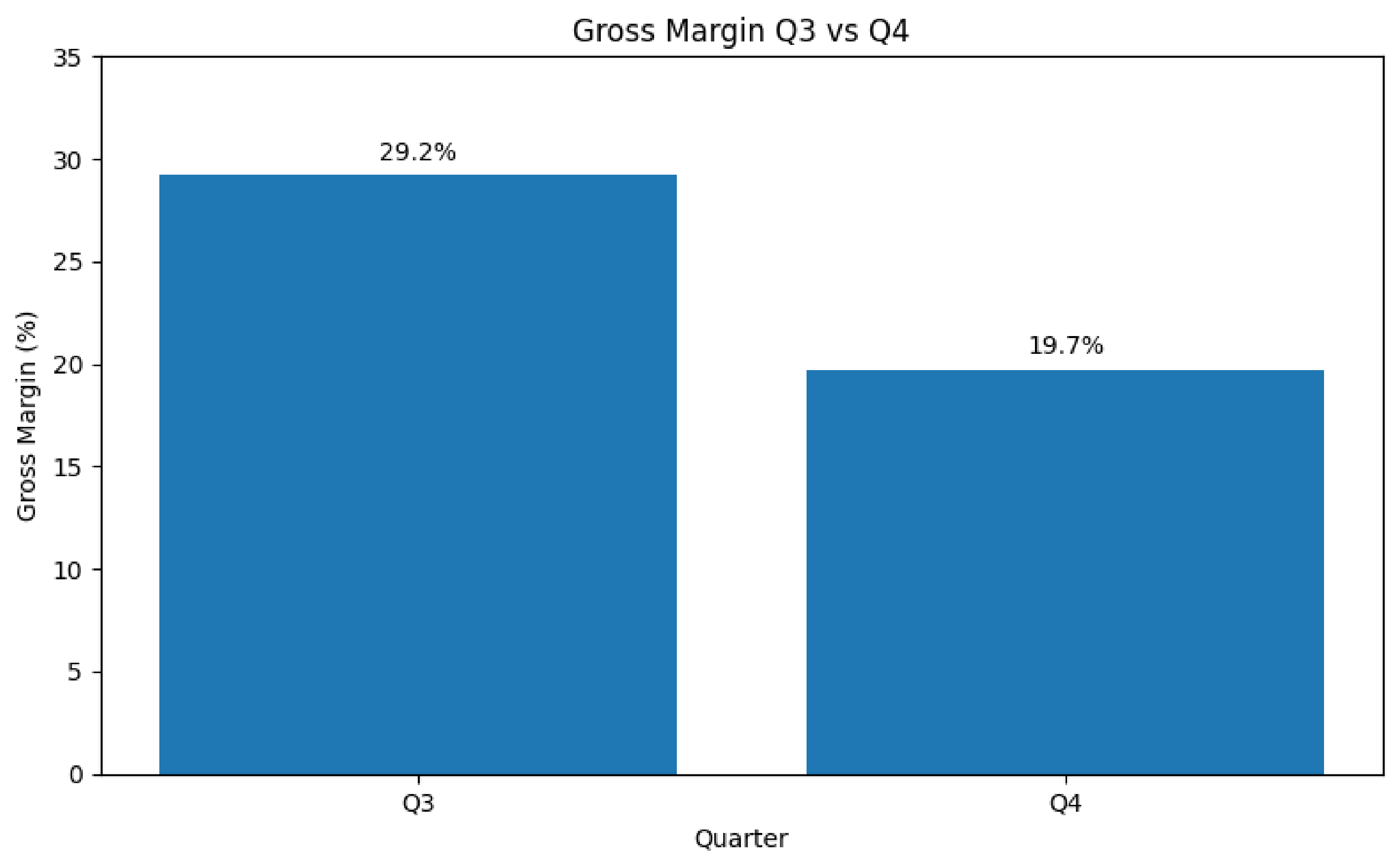
Data Analysis

Questions : **Why did profit decline even though revenue increased?**



Answer:
Revenue increased in Q4, but the profit margin decreased from 29.2% to 19.7%.

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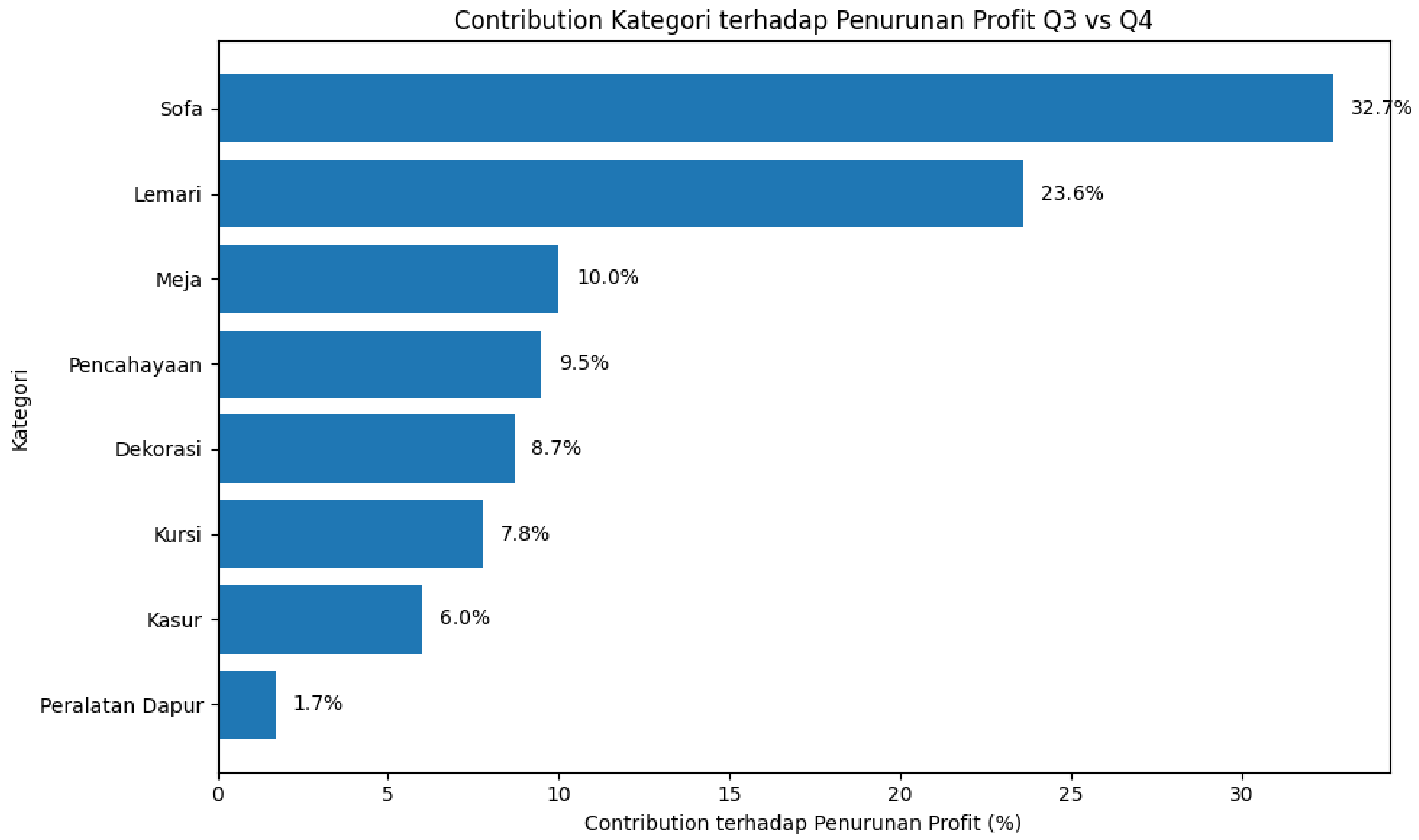
Data Analysis

Questions : **Which category contributed the most to the decline in gross profit?**



Answer:
The decline in Q4 gross profit was mainly driven by the Sofa (32.7%) and Storage Cabinet(23.6%) categories, which together accounted for 56.3% of the total profit decline.

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Visualization

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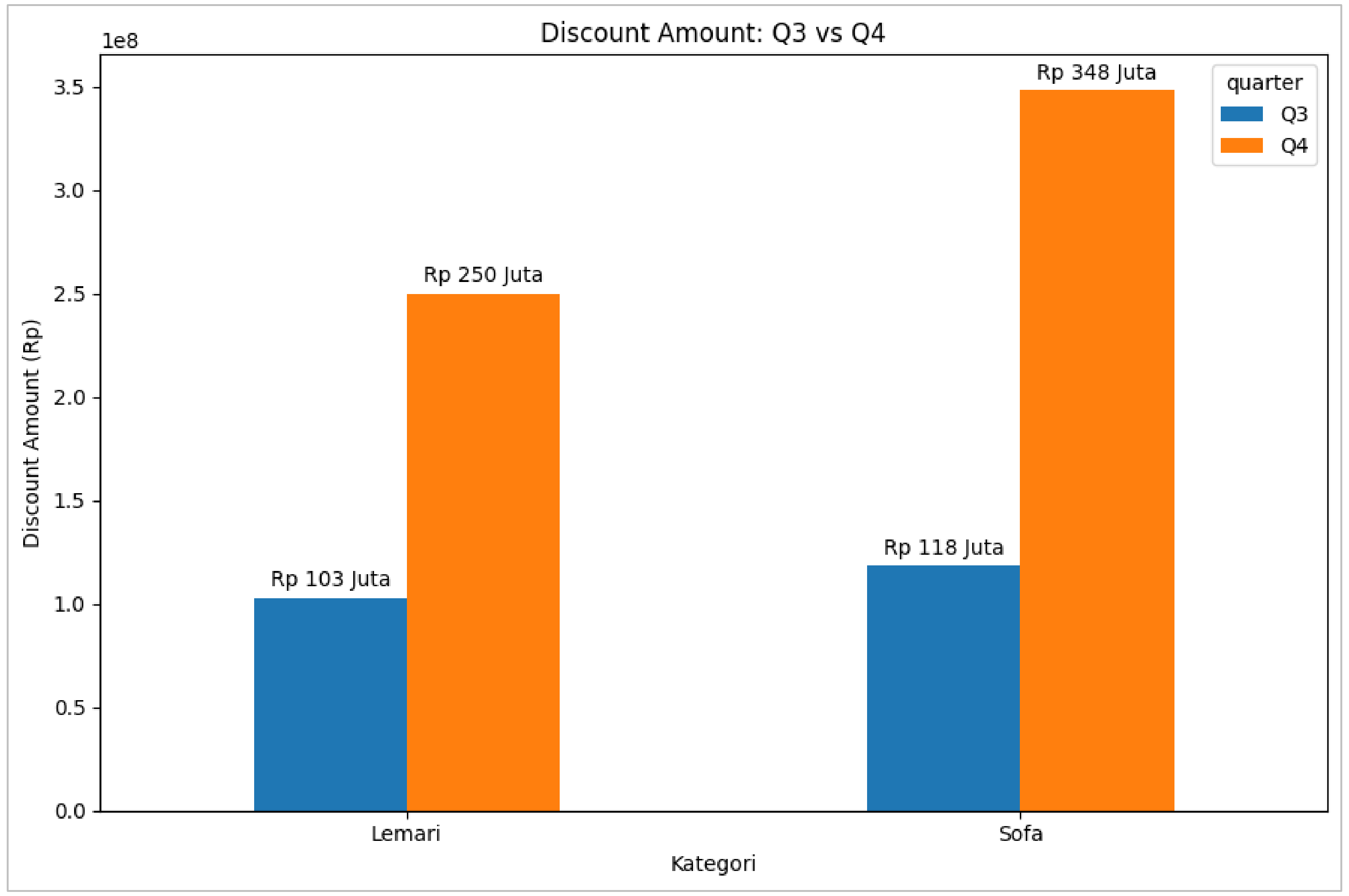
Data Analysis

<u>Questions :</u> What caused the profit decline in the Sofa and Storage Cabinet categories in Q4



Answer: Gross Revenue for Sofa and Storage Cabinet increased in Q4, but higher discounts caused Net Revenue to decline.

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Visualization

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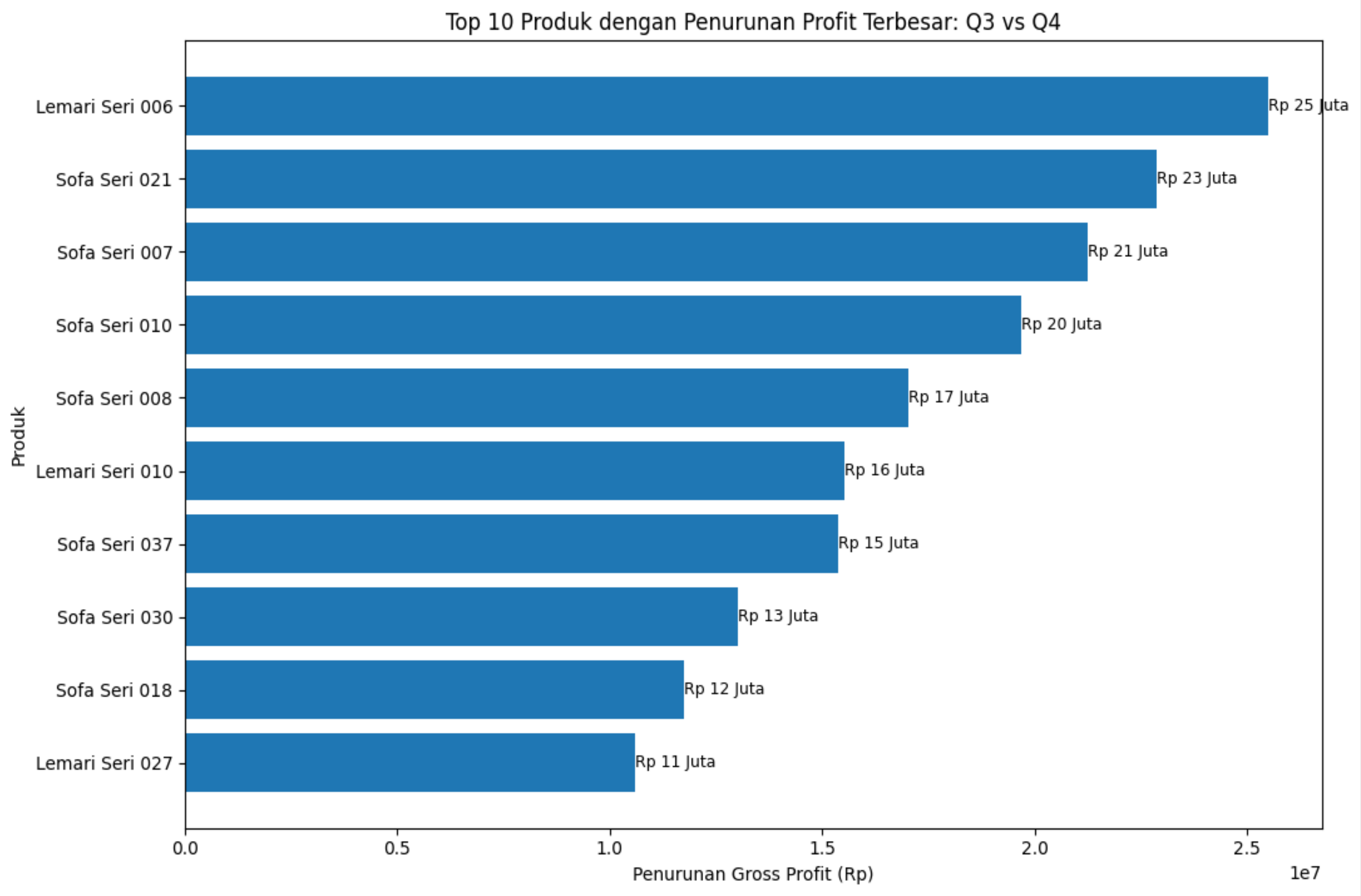
Data Analysis

Questions : Which products contributed the most to the profit decline in the Sofa and Storage Cabinet categories?



Category	Gross Revenue Q3	Gross Revenue Q4	Discount Q3	Discount Q4	Net Revenue Q3	Net Revenue Q4
Storage Cabinet	Rp1.10B	Rp1.22B	Rp102.9M	Rp250.0M	Rp1.00B	Rp965.9M
Sofa	Rp1.39B	Rp1.59B	Rp118.4M	Rp348.1M	Rp1.27B	Rp1.24B

Answer:

The largest profit decline came from several Sofa and Storage Cabinet products, with Storage Cabinet Series 006 being the most affected, declining by approximately Rp25.5 million.

Key Insight

Findings

- Revenue increased, but gross profit declined.
- Gross margin decreased from 29.2% to 19.7% in Q4.
- Sofa and Storage Cabinet accounted for 56.3% of the total profit decline.
- Discounts increased in Q4, putting pressure on net revenue.
- Storage Cabinet Series 006 recorded the largest profit decline, at Rp25.5 million.

Recommendations

Optimize discount strategies for the Sofa and Storage Cabinet categories by reducing discounts on products that put the most pressure on profit, while maintaining promotions for products that still generate healthy margins.

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Debrief

Revenue increased, but profit declined due to margin pressure in Q4. The analysis shows that Sofa and Storage Cabinet were the main contributors to the profit decline, indicating that discount strategies should be selectively optimized for the most affected products.

Conclusion

Key Takeaways

Through this session, I learned the end-to-end data analyst workflow using SQL and Google BigQuery. First, I focused on Data Understanding and Data Profiling by inspecting the dataset, checking the number of rows, date ranges, missing values, invalid values, and duplicate records. Next, I performed Data Transformation and Analysis using SQL by joining transaction and product tables, calculating key metrics such as gross revenue, net revenue, HPP, gross profit, gross margin, and MoM/QoQ changes. I then analyzed the data progressively from monthly and quarterly performance to category and product-level analysis to identify the root causes of profit decline. Finally, I used the SQL results to create Data Visualizations and translate the findings into clear business insights and recommendations. Through this process, I learned how SQL can be used to transform raw transactional data into meaningful insights that support data-driven business decisions.

Let's Connect

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